AN ARTIFICIAL INTELLIGENCE-EQUIPPED SYSTEM FOR THE DIAGNOSIS OF CERVICAL AND BREAST CANCER

Paata Lezhava,
Z.Ghurtskaia
Georgian Technical University

SUMMARY

Discussed the computer system created by us for diagnosing cervical cancer, which is based on artificial intelligence, namely machine learning. The system diagnoses using routine tests and is distinguished by fairly high accuracy. We think the mentioned model will help clinicians identify the tumor and verify the disease in time.

Keywords: machine learning. Diagnosing cancer. model. treatment. Laboratory studies.

ᲮᲔᲚᲝᲕᲜᲣᲠᲘ ᲘᲜᲢᲔᲚᲔᲥᲢᲘᲗ ᲐᲦᲭᲣᲠᲕᲘᲚᲘ ᲡᲘᲛᲡᲘᲕᲜᲔᲔᲑᲘᲡ ᲓᲘᲐᲒᲜᲝᲡᲢᲘᲠᲔᲑᲘᲡᲐᲗᲕᲘᲡ

პაატა ლეჟავა, ზ ღურწკაია
საქართველოს ტექნიკური უნივერსიტეტი

ᲠᲔᲖᲘᲣᲛᲔ

განხილულია ჩვენ მიერ შექმნილი კომპიუტერული სისტემა საშვილოსნოს ყელის სიმსივნის დიაგნოსტირებისთვის, რომელიც დაფუძნებულია ხელოვნურ ინტელექტზე. კერძოდ, მანქანურ დასწავლაზე. სისტემა დიაგნოსტირებას ახდენს რუტინული ტესტების გამოყენებით და გამოირჩევა საკმაოდ მაღაი სიზუსტით. ვფიქრობთ, რომ აღნიშნული მოდელი კლინიცისტებს დაეხმარება სიმსივნის დადგენასა და დაავადების დროულ ვერიფიცირებაში.

The constant mutation and growth of tumors cause changes in blood composition. Doctors have the ability to extract only a small portion of the information hidden in the results of a routine blood test. Scientists at the University of Ljubljana have revealed that more information may be hidden in routine blood tests than even the most experienced sleuth can detect. Using routine blood tests from 15,176 neurological patients, they created a predictive machine-learning model for diagnosing brain tumors. The sensitivity and specificity of the adapted tumor model were 96% and 74%, respectively. Our research aims to create an artificial intelligence-based diagnostic model for cervical and endometrial tumors using patient databases.

Therefore, a more serious and accurate analysis of the results of routine blood tests is important, because we can find evidence of specific tumor growth in the body. Today, scientists are increasingly using artificial intelligence methods to solve this type of problem. Machine learning, which is widely used in artificial intelligence, can reveal such patterns from medical data that are impossible even for highly qualified human doctors. Such successful diagnostic models will greatly assist clinicians in timely diagnosis verification.

The diagnosis of cervical, endometrial breast tumors is usually based on the patient's anamnesis, instrumental and laboratory studies. Early diagnosis and timely treatment of cancer is important, which in many cases is not possible due to the absence of symptoms and a late visit to the doctor. Non-specific clinical manifestations and relatively rare forms of tumors are also a problem. Although computed tomography and magnetic resonance imaging studies will undoubtedly be important. Doctors are able to extract only a fraction of the information hidden in routine blood test results. Therefore, a more serious and accurate analysis of the results of routine blood tests is important, because we can find evidence of specific tumor growth in the body [2].

Today, scientists are increasingly using artificial intelligence methods to solve this type of problem. Machine learning, which is widely used in artificial intelligence, can reveal such patterns from medical data that are impossible even for highly qualified human doctors. Such successful diagnostic models will greatly assist clinicians in timely diagnosis verification [1].

As a result of retrospective analysis of patient histories, it is possible to create a diagnostic model based on machine learning, which is gaining more and more interest today [4].

RESEARCH DESIGN

We studied the history of oncology patients who were diagnosed with cervical and endometrial cancer and underwent treatment at St. Tbilisi Experimental and Research National Center Of Surgery. The history of 344 patients was studied. We used only routine laboratory studies to construct the model (Table.1)

(Table.1)

TABLE

1.General blood analysis
leukocyte (WBC)
erythrocytes (RBC)
hemoglobin (HGB)
Hematocrit (HCT)
Mean erythrocyte weight (MCV)
Hemoglobin avg. Content in erythrocyte (MCH)
Hemoglobin avg. Concentration in erythrocyte (MCHC)
erythrocyte distribution area (RDW-CV)
Platelet (PLT)
Mean Platelet Volume (MPV)
Neutrophil % (neutro)
neutrophil
lymphocyte % (LYMPH%)
lymphocyte
monocyte % (MONO%)
monocyte
Eosinophil % (eosino %)
Eosinophil
Basophil % (BASO%)
Basophil
Eds- Westergreen method

2. Analysis of liver functions
Alkaline phosphatase concentration determination
Determination of alanine aminotransferase concentration
Determination of aspartate aminotransferase concentration
Determination of direct bilirubin
Determination of total bilirubin
Determination of creatinine concentration in blood
Determination of gammaglutamyltranspeptidase concentration

3. Diagnosis of infectious diseases in patients
HBsAg (rapid test)
HIV-1/2
RPR (syphilis test)

4. Determination of fasting glucose concentration (GLUC)

5. Determination of blood group and rhesus

6. Coagulogram
FIBR
PT
PI %
INR
TT
aPTT

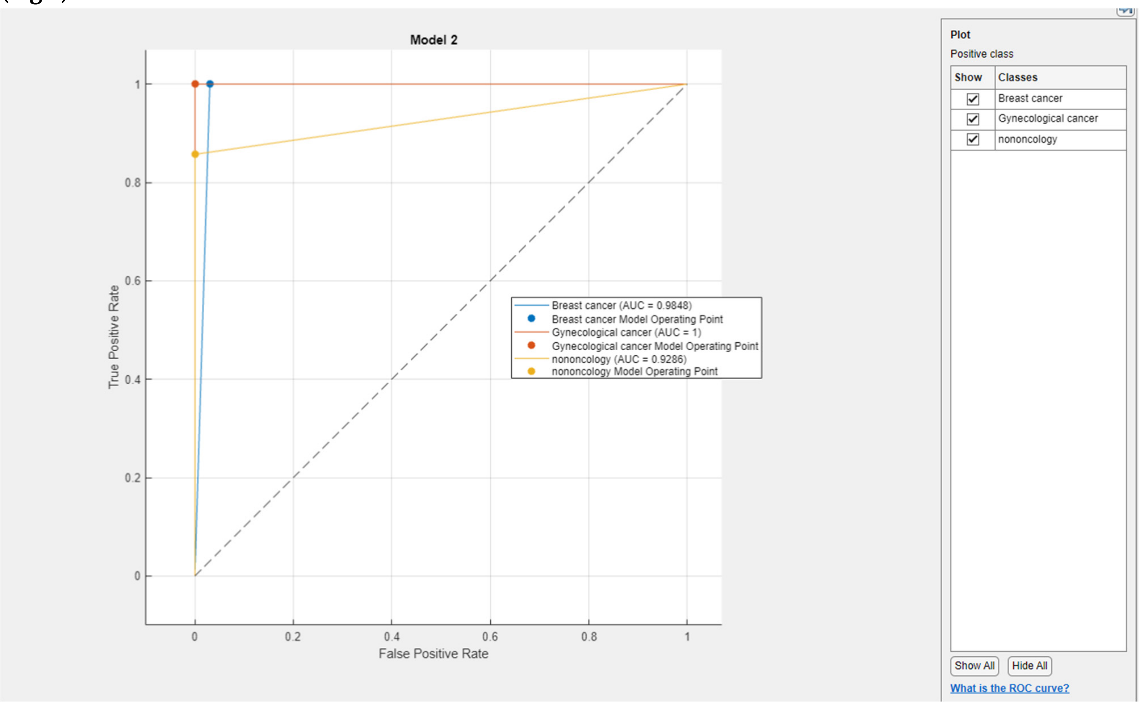
7. General analysis of urine
quantity
Specific weight
reaction
Color, transparency
protein quantity
Flat epithelium
The erythrocyte is irreplaceable
leukocyte
bacteria
salts

To build the model, we used a supervised machine learning method, namely the classification, that best suits our task. Matlab software was used as a tool, namely classification learner [1].

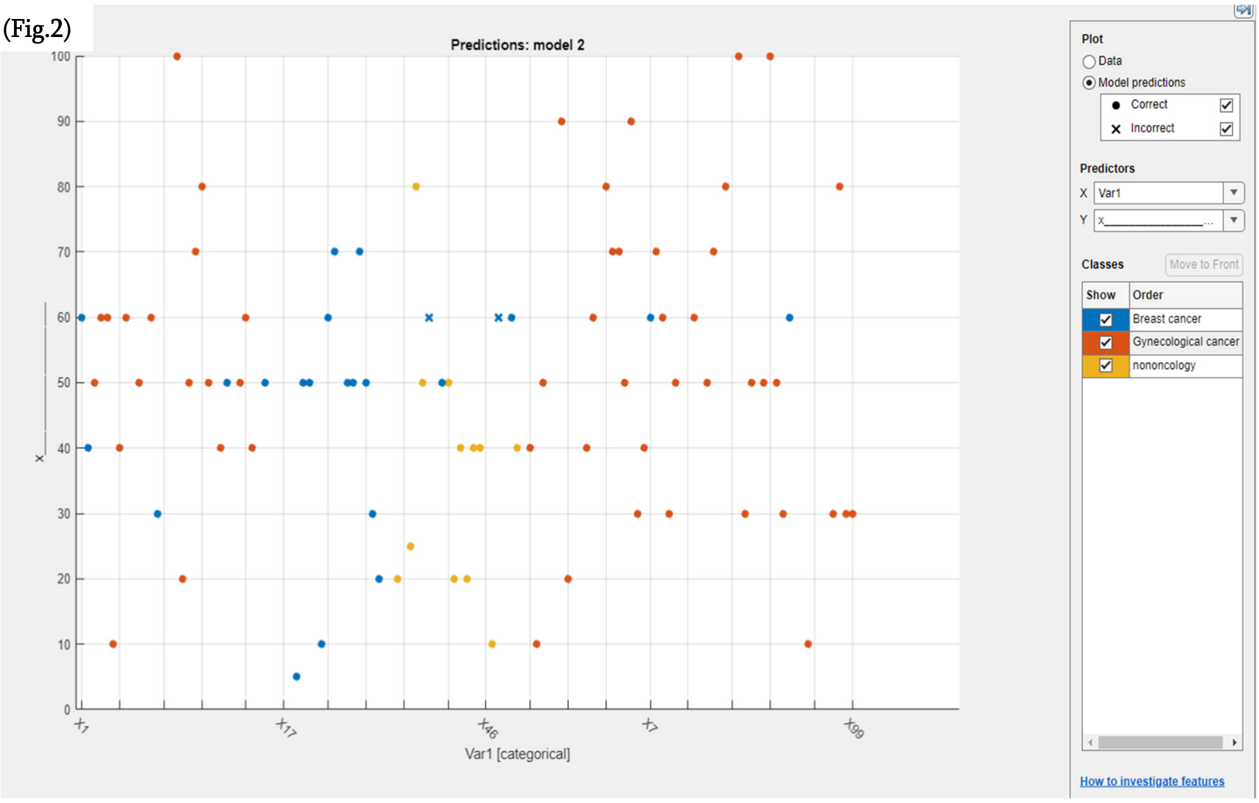
The model was trained using the following statistical methods. See the pictures for the corresponding results. Ensemble, Boosted Tree, fine Tree, linear SVM, Quadratic SVM, neural network, Cubic SVM, logistic Kernel, SVM kernel, Coarse tree, Medium Tree.

The best results were obtained using the Medium Trees method (Fig.1,2,3,4,5). It identified patients with cervical tumors with 100% accuracy, noncancer patients with 90% accuracy, and breast cancer patients with 93.3% accuracy.

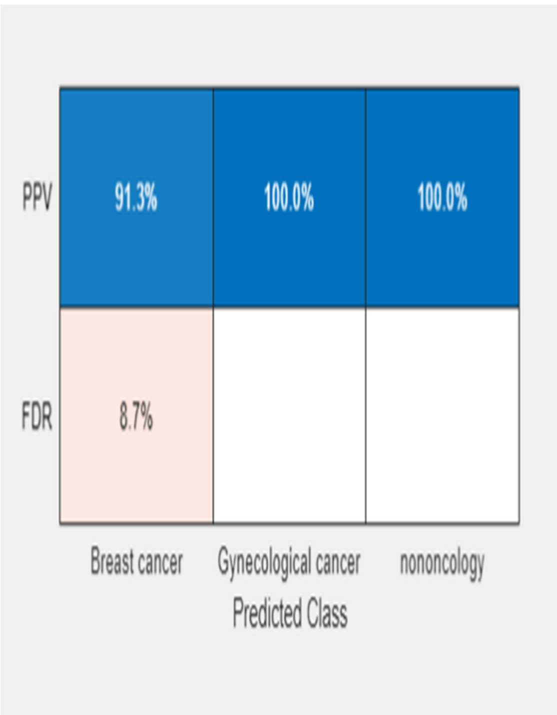
(Fig.1)



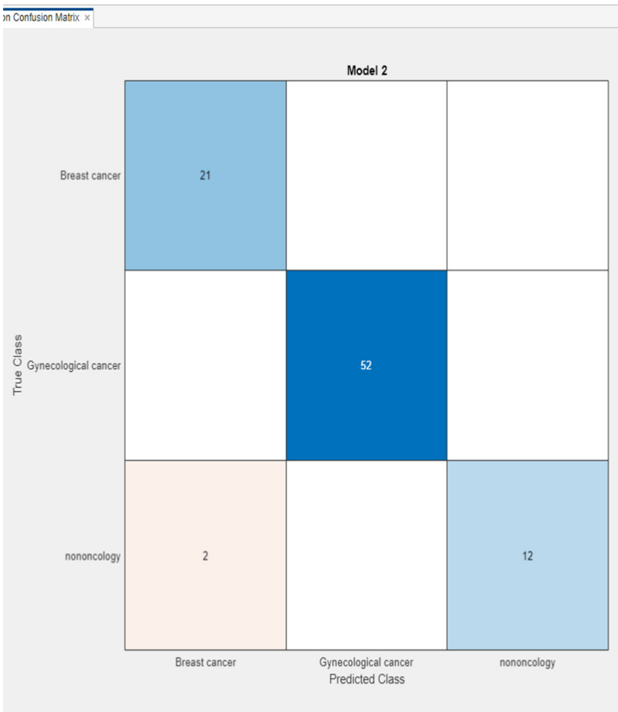
(Fig.2)



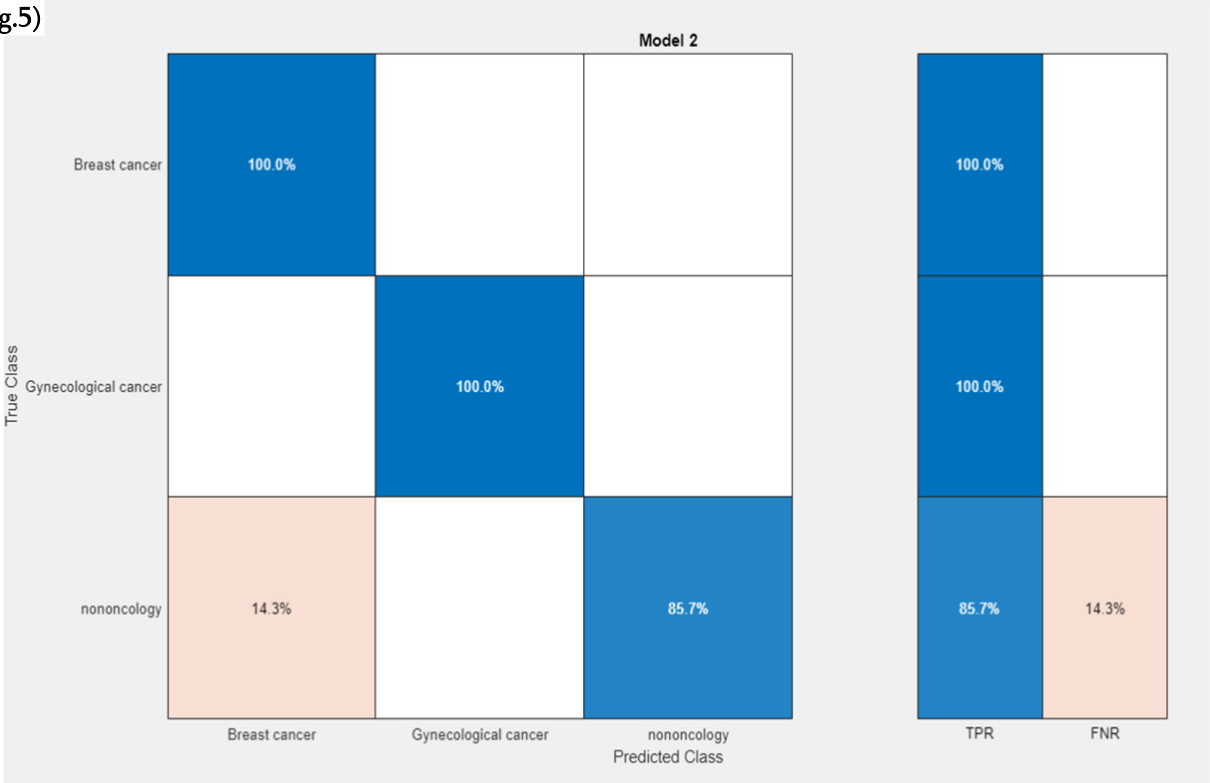
(Fig.3)



(fig.4)



(fig.5)



CONCLUSION

A machine learning-based computer system for cervical and breast tumor diagnosis has been created, which diagnoses using routine tests and is distinguished by fairly high accuracy. Patients of the Georgian population, who were treated at the National Center of Surgery, were used for teaching the system. We think this model will provide important help to clinicians in timely verifying cervical and breast tumors.

LITERATURE:

1. Natalie Sidamonidze(2019) Artificial intelligence as a challenge and some methodological aspects of its realization. Georgian Technical University. AUTOMATED CONTROL SYSTEMS - No 1(28), 2019 186.

2. Simon Podnar, corresponding author Matjaž Kukar, Gregor Gunčar, Mateja Notar, Nina Gošnjak, and Marko Notar (2019). Diagnosing brain tumors by routine blood tests using machine learning. doi: 10.1038/s41598-019-51147-3

3. Torlay L, Perrone-Bertolotti M, Thomas E, Baciu M. Machine learning-XGBoost analysis of language networks to classify patients with epilepsy. Brain Inform. 2017;4(3):159–69. doi 10.1007/s40708-017-0065-7.

4. Mariam Chkhaidze, Otar Tavdishvili, Giorgi Chichua, Sofia Barnov (2020). Artificial intelligence (methodical guidelines for practical work). Georgian Technical University. (in Georgian)